

Sebastian Ashcallay Silva

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Software Engineer specializing in AI and multimodal pipelines, experienced in production-ready systems, distributed workflows, and driving measurable user impact.

EDUCATION

University of Maryland

Bachelor of Science in Computer Science (ML Concentration), Minor in Mathematics

College Park, MD

Jan. 2024 – Dec. 2025

EXPERIENCE

Software Developer Intern

Apollonian AI

Jan. 2025 – Present

Remote

- Led development of a production-grade multimodal video generation pipeline, integrating LLM-based script generation, image synthesis, TTS audio, and automated rendering, supporting 5,000+ mentor conversations and 350+ daily active visits
- Architected backend services using Python and Firebase Functions (GCP), managing API-driven workflows, Firestore data models, and cloud-based asset storage for scalable media generation
- Implemented asynchronous task orchestration using batched API requests and serverless triggers, reducing end-to-end video generation runtime by 50% (6 min → 3 min)
- Applied prompt engineering to integrate LLM and multimodal AI services into a unified pipeline
- Developed internal testing and validation tools (HTML/CSS/JS) adopted by team members to debug the multi-stage process, improving reliability and reducing iteration time
- Owned end-to-end system design and deployment workflows, including FFmpeg-based video processing on cloud instances with temporary storage and automated asset delivery via Firebase
- Collaborated with CEO on product direction and go-to-market strategy, contributing to pitch decks, technical positioning, and outreach to university networks, driving 120+ monthly signups and 10+ premium conversions
- Represented company in university-backed startup competitions (Pitch Dingman, XFoundry Mental Health Xperience at UMD), contributing to funding pitches, partnership development, and early user acquisition strategy

Tutor

C2 Education

Sep. 2024 – Jul. 2025

Bethesda, MD

- Led 15+ structured 2-hour weekly sessions for K-12 school students in Math, Physics, Economics and Computer Science, improving performance in SAT/ACT/IB exams and problem-solving ability
- Increased SAT Math scores from 580–620 range to 700+ through personalized learning plans

PROJECTS

Professor Ratings Predictor | *Python, Pandas, Scikit-Learn, NLTK*

Feb. 2025 – Apr. 2025

- Built end-to-end ML workflow on 37,000+ reviews across 4,100+ professors, transforming unstructured text into structured predictive features
- Engineered sentiment analysis features using NLTK, significantly improving correlation with rating outcomes
- Trained and evaluated regression models (Ridge, SVR, Gradient Boosting) with 10-fold cross-validation, achieving RMSE 0.452 and R^2 0.785
- Performed feature importance analysis and dimensionality reduction (PCA), generating visual insights from student feedback data

City Classification via CNNs | *Python, PyTorch, ResNet18*

Mar. 2025 – May 2025

- Built CNN-based classification system using transfer learning with ResNet18 to identify U.S. cities from sky projection images, achieving 95.25% precision and 100% Top-3 accuracy
- Designed preprocessing and augmentation pipelines (noise, blur, light pollution), creating 2 datasets with 2,700+ images to improve robustness
- Implemented full training and evaluation workflow using PyTorch and Scikit-Learn, including normalization, F1 scores, and confusion matrix analysis
- Co-authored a 5-page report summarizing methodology, experiments, and model performance insights

CERTIFICATIONS

AWS Cloud Practitioner

Amazon Web Services

Issued Feb. 2026

TECHNICAL SKILLS

Languages: Python, JavaScript, SQL, Java, C/C++, HTML/CSS

AI / Machine Learning: LLMs & multimodal AI pipelines (GPT-4o, image generation, Whisper), PyTorch, Scikit-Learn, NLTK, Prompt Engineering, Model Evaluation

Frameworks & Tools: React, Node.js, Express.js, FastAPI, Flutter, Firebase (Cloud Functions, Firestore, Storage), REST APIs, Serverless, Asynchronous & Distributed Workflows, Fault Tolerance (Retries, Rate Limiting)

Data & Infrastructure: Pandas, NumPy, JSON & NoSQL Schema Design, Base64 Encoding/Decoding, FFmpeg, Environment & Secrets Management, Git, Bash, GCP, AWS

Other: Microsoft Office (Word, Outlook, PowerPoint, Excel), Figma